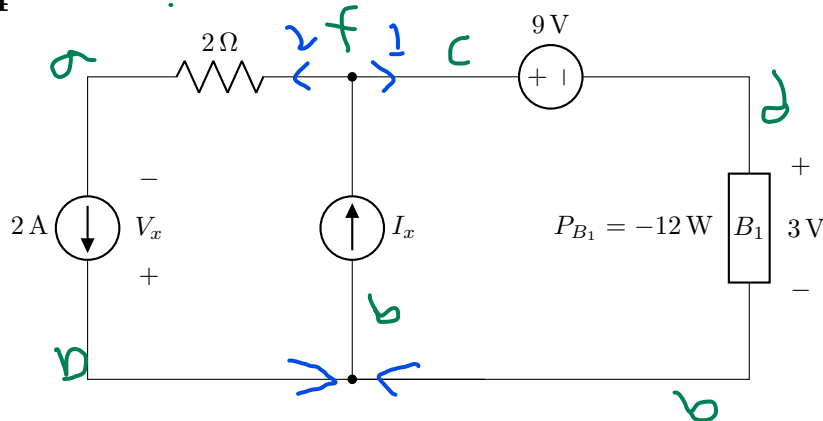


- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ All **4 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

■ Question 1 of 4

[CO3] [8 marks]

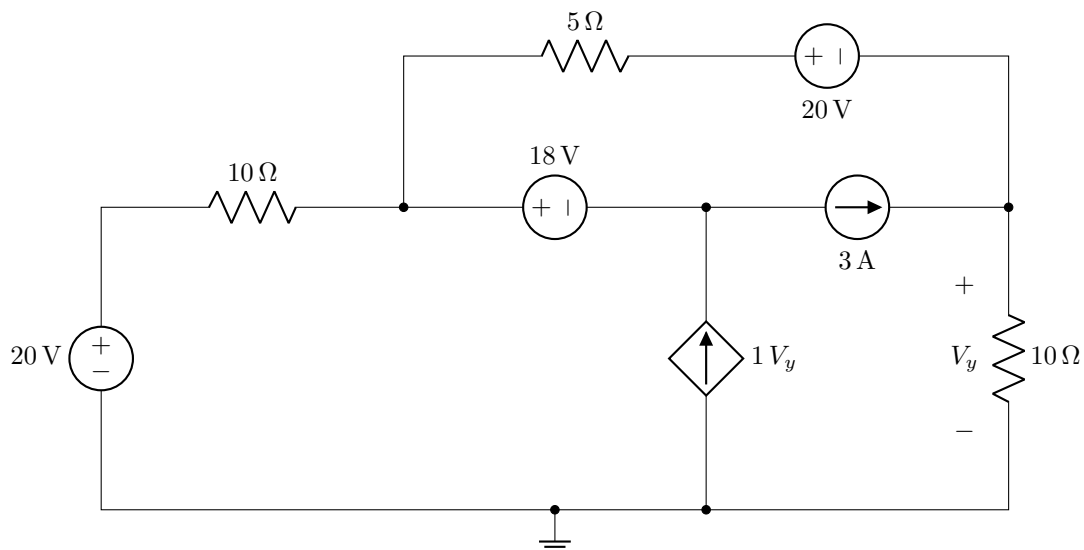


Apply **KCL**, **KVL**, and **Ohm's Law** to answer the following questions–

- (a) [3 marks] Determine the current through the 9 V voltage source if the power of the element B_1 is -12 W .
- (b) [3 marks] Determine I_x .
- (c) [2 marks] Determine V_x , the voltage across the 2 A current source.

■ Question 2 of 4

[CO3] [24 marks]



For the circuit shown above,

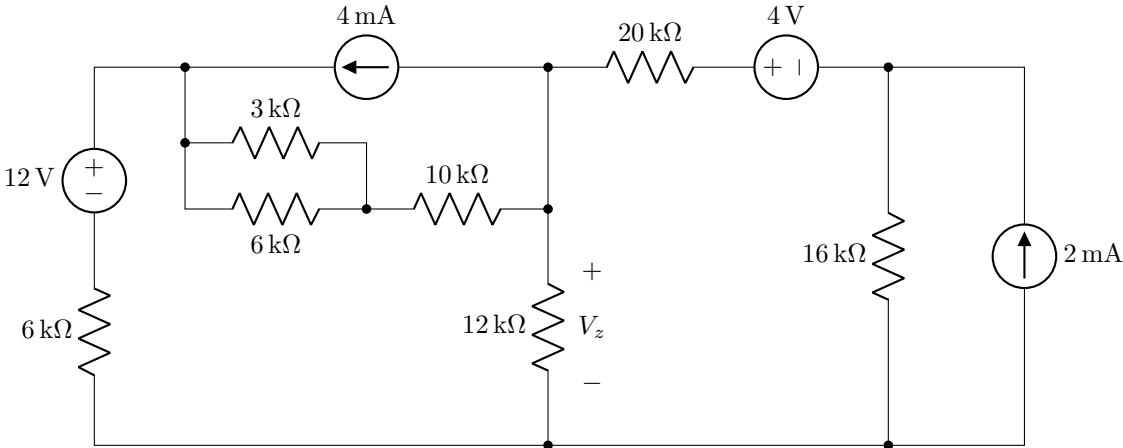
- (a) [16 marks] **Apply** either **Nodal Analysis** or **Mesh Analysis** to determine the power of (i) the 18 V voltage source and (ii) the 3 A current source with the appropriate $\pm$ sign and units.^{††} Also, mention in each case whether the source is supplying or absorbing power.
- (b) [8 marks] **Apply** the alternative method that you did not use in (a) to formulate all the equations needed to solve the circuit. You do not need to simplify or solve the equations.

^{††}Node voltage or mesh current variables must be labeled on the diagram

■ Question 3 of 4

[CO2] [14 marks]

Apply **Source Transformation** to reduce the following circuit to a single loop and then determine V_z .



■ Question 4 of 4

[CO3] [8 marks]

Determine the equivalent resistance between the outermost nodes of the Resistobot's hands, that is, between the circles in the given figure.

